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Waves or ripples: African countries risk shocks and the survival of Chinese exports to Africa

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ABSTRACT

This paper examines the impact of risk shocks in Africa on the stability of Chinese firms' export relationships by analyzing risk fluctuations and using survival analysis to measure their effect. The findings reveal that risk shocks significantly increase the survival risk of Chinese firms' export relationships, leading to a notable reduction in stability. Private and non-foreign firms are more vulnerable to these risks, but certain market strategies such as learning-by-doing, learning from neighboring firms, and waiting strategies can effectively mitigate their adverse effects on export stability. Our findings indicate that certain market strategies, such as learning-by-doing, learning from neighboring firms, and waiting strategies, can effectively mitigate the adverse effects of risk shocks on export stability. Interestingly, a nonlinear marginal effect is observed in the case of the waiting-timing strategy. However, adopting a multi-destination strategy tends to worsen the negative impact of risk shocks on export stability, especially for private firms. Furthermore, non-market strategies employed by firms that promote government diplomatic behavior help reduce this negative impact.

1. Introduction

The risk of African countries has long been a major hindrance to economic cooperation with Africa for governments and firms alike. Negative perceptions, such as a 'poor business environment', 'terrorism', and 'political turmoil', disssuade companies from entering African markets. High country risk is haracterized by inconsistent foreign economic policies, leading to increased government default risk, supply chain market risk, and customer credit risk for firms. Particularly noteworthy are the unpredictable fluctuations in risks in African countries, stemming from events like partisan disputes, terrorist activities, epidemics, tribal conflicts, and religious tensions. While existing studies often emphasize the high overall risk levels in African countries as a deterrent for firms to export to Africa, few have delved into the impact of sudden risk fluctuations on bilateral trade cooperation. This study takes a dynamic approach to identify risk shocks in African countries and assess their economic effects on Chinese firms' exports to Africa. Additionally, the study aims to propose strategies that can assist firms in better managing these risk shocks.

China has historically been a significant trading partner for Africa, with the Belt and Road Initiative further solidifying this relationship since its inception in 2013. The trade between the two regions has gained momentum, especially as more private firms look to tap into African markets (China-Africa Business Council, 2022). However, entering emerging markets like Africa comes with

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increased uncertainty. The risk shocks faced by African countries pose a challenge for firms looking to expand into these markets (Yan & Luis, 2013). In Fig. 1, while the number of Chinese enterprises exporting to Africa has grown, there is a concerning pattern of high turnover among these firms annually.

It is crucial to not only increase the number of exporting firms but also to cultivate and enhance export relationships. The challenges faced in Chinese export relationships with Africa are highlighted in Fig. 2, which depicts a steady rise in new export relationships alongside a growing number of export relationships exited. This trend emphasizes the need to address the unique risks and market dynamics in Africa, that contribute to the high entry and exit rates of Chinese export enterprises in the region.

Moreover, maintaining trade stability with African countries poses a significant challenge for Chinese firms. The current export relations of Chinese firms to Africa are characterized by a high rate of exit and short duration of survival. This means that many firms enter and exit the African export market rapidly, which may have a detrimental impact on their export confidence and preference for African markets. Several studies suggest that enhancing the stability of new export relations is crucial for achieving export growth (Brenton et al., 2010). Therefore, it is important to improve the survival rate of firms in exports and ensure the stability of export relationships in order to promote export growth (Monarch & Schmidt-Eisenlohr, 2023). Specifically, fostering the long-term survival of emerging export firms is crucial to deepen China-Africa trade cooperation. This article examines how Chinese firms' export relationships withstand risk shocks in African countries and proposes strategies to bolster their resilience against risks. Thus, this paper contributes to the existing literature on the tension between China's foreign export growth and the impact of risks in destination countries.

Compared with existing literature, this paper makes marginal contribution in three aspects. Firstly, it broadens the research perspective on risk issues related to African countries. Previous studies have mainly focused on overall difference in risk levels among African countries from a static perspective. However, this paper emphasizes the influence of risk volatility characteristics in destination countries from a dynamic perspective. Additionally, it constructs an index measuring African risk shocks by identifying abnormal fluctuations in risk volatility across African countries. Secondly, this paper utilizes comprehensive dataset comprising Chinese firm-level exports data at both the firm level. It analyzes the stability of Chinese firms' exports to Africa and investigates the economic impacts of risk shocks in African countries on export stability using the survival analysis method. Thirdly, the paper examines the strategies employed by firms to enhance export stability and mitigate risks through market and non-market strategies. The findings reveal that firms' market strategies have heterogeneous moderating effects on the impact of risk shocks on export stability, depending on their characteristics. Moreover, firms' non-market strategies can reduce the negative impact of risk shocks on export stability by promoting government diplomatic behavior.

The remainder of the paper is organized as follows: Section 2 provides a comprehensive review of the relevant literature and presents our research hypotheses. In Section 3, we provide detailed information on the two datasets and present the measurement results of the core indicator. Section 4 introduces the empirical method and provides in-depth descriptive statistics. The main empirical findings, along with several robustness checks, are presented in Section 5. We thoroughly discuss the impacts of firms' export strategies

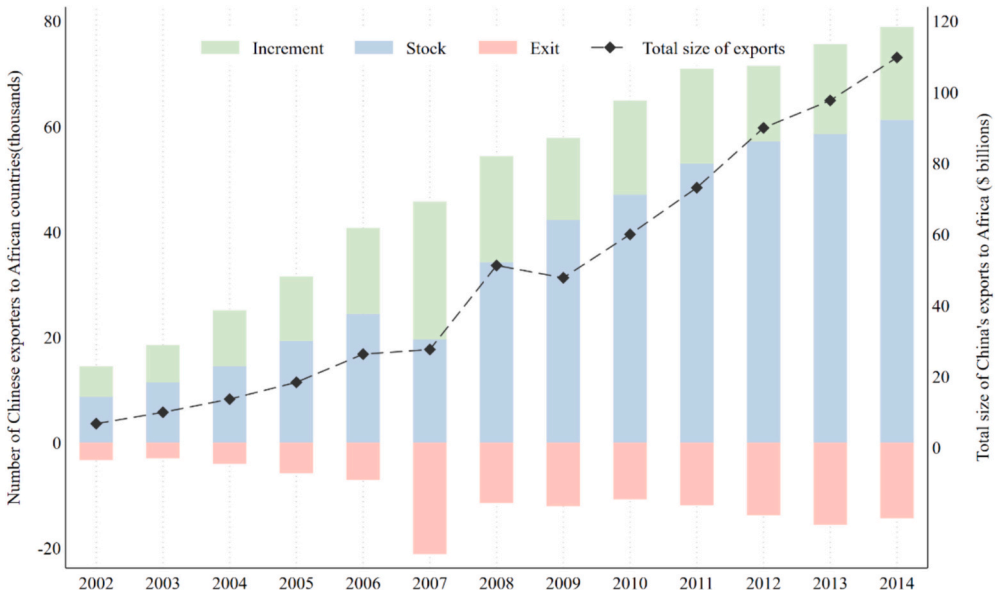


Fig. 1. Changes in the scale of China's exports to Africa and the number of surviving firms.
Note: (i) The data presented in this study is based on China's customs data. (ii) The number of exits of Chinese firms exporting to Africa each year is estimated by considering the stock of the current year, the stock of the previous year, and the number of new firms added in the current year. The calculation is done using the formula exit_f = l.stock_f + entry_f - stock_f. (iii) The new firms added are only those considered as first-time exporters to Africa and do not include intermittent exporters. While this calculation may underestimate the number of exits from China's exports to Africa each year, it does not impact the overall conclusions drawn from the data.

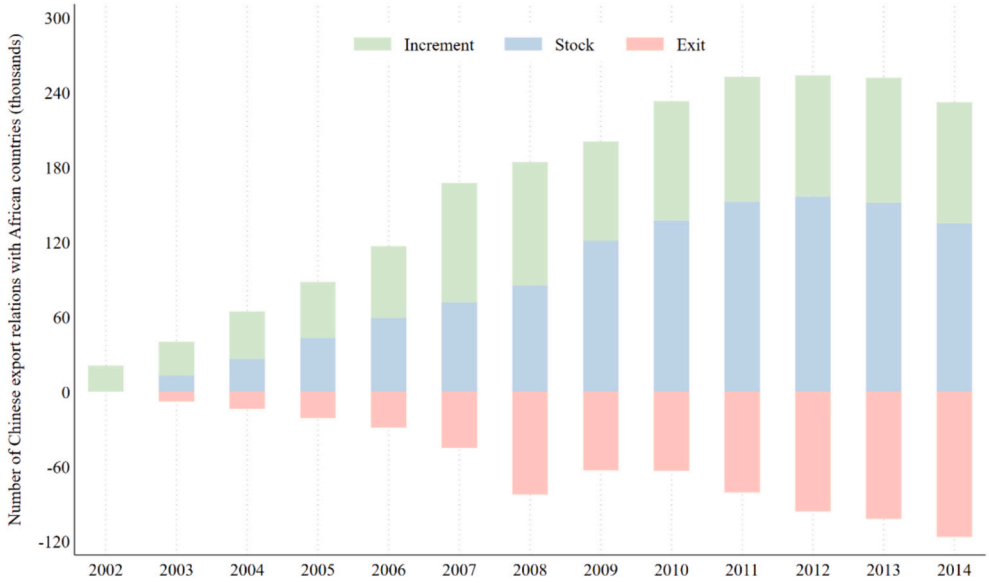


Fig. 2. The number of existing export relationships at the firm-country level.
Note: (i) The results presented in the figure are derived from China's customs data. (ii) In this Fig. 2, the addition and withdrawal of export relationships are simplified to refer to the first time an enterprise exports to a country (Increment) and when the enterprise no longer exports to that country within the study timeframe (Exit). More detailed definitions and illustrations are provided in Section 3.2.

on heterogeneity moderating influences in Section 6 and offer conclusions in Section 7.

2. Literature review

2.1. Risk and firms' export stability

The phenomenon of unstable export relationships among firms is common in international trade, and numerous studies have shown that these relationships tend to be short-lived and negatively impacted by the passage of time, based on trade data from various countries (Besedes & Prusa, 2006; Shao et al., 2012; Straume, 2017). Additionally, several studies have examined the factors influencing the duration of firms' export relationships (Mayneris & Poncet, 2015). This paper is closely related to the work conducted by Esteve-Pérez et al. (2013), who empirically established that the level of political risk in a country plays a significant role in constraining the duration of firms' export relationships. Based on their findings, this study highlights the negative impact of risk on the survival of a firm's export relationship which can be categorized into two aspects. Firstly, it emphasizes that throughout the exporting process, firms need to continuously adapt their decision-making process regarding the overall risk level associated with exporting to specific destinations. Vannoorenberghe (2014) proposes that the overall risk level is often perceived as an integral component contributing the fixed costs associated with exporting while significantly influencing firms' export decisions. Secondly, African countries stand out due to their higher-than-average volatility in terms of risks arising from conflicts, epidemics, and other unpredictable issues. These risk shocks pertaining to risks within destination market gradually become a significant concern that undermines export stability.

Firms face higher fixed costs for exporting and increased uncertainty in trade policy due to risk shocks in the destination market. To mitigate potential risks during the export process, exporters are required to conduct preliminary investigations of the destination market and obtain export credit insurance (Baltensperger & Herger, 2009; Choi & Kim, 2021). However, local risk shocks lead to increased costs for market research and export credit insurance, thereby weakening firms' operational capacity and reducing their risk-taking ability. Furthermore, previous studies have shown that demand uncertainty and policy uncertainty in destination markets influence firms' decisions on whether to enter or exit (Nguyen, 2012; Handley & Limo, 2017; Caldara et al., 2020; De & Gaigné, 2020). Therefore, we argue that risk shocks in the destination country lead to a greater mismatch of information in the export market, potentially causing firms to suspend or terminate exports to that particular market. This phenomenon helps to elucidate the high exit rate and instability of Chinese firms' export relationships with Africa. Therefore, we propose the following research hypothesis:

Hypothesis 1. Risk shocks in the destination country decrease the duration of Chinese firms' export relationships with Africa.

2.2. Export strategy and export stability

Profit-driven motives drive compel firms to continuously explore new export markets. The multi-market export strategy yields positive effects, primarily through the scale effect and information spillover effect. By expanding their exports, firms can benefit from economies of scale, leading to reduced marginal costs of production. Additionally, there is an information spillover effect resulting

from the complementary knowledge acquired in multiple markets. For multi-market exporters, knowledge gained in one export market enhances risk-warning capabilities and facilitates faster acquisition of new knowledge from other markets (Mayneris & Poncet, 2015). However, firms engaged in multi-market exports often face higher fixed costs and business risks, particularly during the initial stages of establishing export relationships for new exporters. There are also negative consequences associated with the multi-market export strategy. Firstly, it increases sunk costs intensifying financial pressure on firms. Secondly, it raises the likelihood of firms encountering risk shocks due to the contagious nature of risks between countries.

The impact of multi-market strategies on the overall export survival of firms and the export relationships between firms and individual destination country markets can vary. The export stability of firms depends on firm size and the diversification of their export markets. Export market diversification is beneficial for large-scale firms as it enhances their ability to withstand risks. However, for small traders, export market diversification may lead to increased volatility in exports (Vannoorenberghe et al., 2016). In terms of export relationships, firms tend to accumulate experience by initially establishing export relationships with markets that have lower trade policy uncertainty. They then gradually enter markets with higher levels of trade policy uncertainty to improve the survival rate of export relationships (Sheard, 2014). This paper aims to investigate whether a multi-market strategy enhances the risk-resistant ability of firm-destination level relationships in China-Africa trade. Therefore, this paper proposes research hypothesis 2a.

Hypothesis 2a. Multi-market strategy exacerbates the negative impact of risk shocks on the survival time of export relationships.

Several studies have demonstrated that prior exporting experience positively influences firm survival rates during the establishment of new export relationships (Albornoz et al., 2016). When faced with information asymmetry in entering unfamiliar markets, companies encounter substantial sunk costs. By employing experiential-based learning strategies derived from past encounters (Love & Manez, 2019; Esteve-Pérez, 2021). However, the decision to export is an ongoing learning process, and firms can continuously learn and adjust through strategies such as the stem-learning effect. This involves summarizing and learning from their own export experience. In practice, firms can accumulate export experience by starting with small orders and gradually exploring the market to understand its demands (Berman et al., 2019; Eaton et al., 2021). Another learning strategy is the peer effect, where firms learn from the experiences of first movers (Fernandes & Tang, 2014). Additionally, firms can improve trade stability by learning from neighboring firms and making joint export decisions (Fabling & Sanderson, 2013; Wagner & Zahler, 2015). Based on these points, this paper proposes research hypothesis 2b.

Hypothesis 2b. Learning strategies mitigate the negative impact of risk shocks on the survival time of firms' export relationships.

The timing of a firm's wait strategy has option value, as it entails a trade-off between mitigating potential risk losses and capitalizing on potential export gains (Handley & Limo, 2017). Firms adjust the forecast interval of their export earnings b in response to changes in trade policy uncertainty. If a firm's expectation of future export earnings experiences a significant downgrade or becomes negative, it may consider delaying exports as a risk mitigation measure (Nguyen, 2012). Moreover, longer durations of uncertainty and higher productivity thresholds will lead to longer waiting times for exporting (Duan, 2023). Therefore, this paper proposes research hypothesis 2c.

Hypothesis 2c. The wait-to-re-export strategy has a nonlinear moderating effect on the relationship between risk shocks and firms' export survival time in export relationships.

2.3. Government diplomatic behavior and export stability

Many studies have demonstrated that improving bilateral political relations and international linkages promotes trade cooperation between countries, while conflicts in bilateral political relations diminish the extent of bilateral trade (Du et al., 2017; Heilmann, 2016; Golley et al., 2022; Le, 2022). Diplomatic behaviors, such as high-level visits between China and African countries, play a crucial role in enhancing bilateral political relations and trade. State visits, in particular, contribute to increased official aid to Africa and exports from state-owned firms (Lin et al., 2017). Resolving disputes on bilateral political issues is identified as the primary means to prevent the deterioration of bilateral political relations through diplomatic negotiations (Bell & Long, 2016). Furthermore, government diplomatic behavior can facilitate the conclusion of trade agreements, creating a favorable trade environment (Plouffe & Van, 2016). For export firms, a stable trade policy environment effectively safeguards their legitimate rights and incentivizes them to handle shocks. Additionally, bilateral political relations influence bilateral trade flows by shaping the national image identity of consumers in the destination country, thereby impacting their consumption preferences (Pandya & Venkatesan, 2016; Rose, 2016). Based on these findings, this paper proposes a third research hypothesis.

Hypothesis 3. The government's bilateral diplomatic behavior helps mitigate the negative impact of home-risk shocks on the duration of firms' export relationships.

3. Indicator measurement and typical facts

3.1. Measurement of risk shocks

Country risk fluctuation in export destinations are a crucial determinant impacting the sustainability of firms' export relationships. Additionally, there is a threshold effect for country risk fluctuation on the stability of these export relationships. However, data related to the overall level of country risk or economic uncertainty[1] has limitations, as it fails to capture the nonlinear, abrupt, and heterogeneous characteristics of the impact of country risk on firms' export stability. Therefore, we use concepts from Forbes & Warnock (2021) as a basis for defining and calculating indicators for risk shock in African countries. Specifically, risk shock is defined as an abnormal fluctuation of the risk level beyond the scope of the national system of the destination country. The raw data utilized in this study is sourced from the ICRG (International Country Risk Guide) database,[2] which covers 35 African countries.[3] The specific processing steps are as follows.

(1) We assess the level of risk by calculating the moving sum of country risk level for each African country every three months: $C_t^{risk} = \sum_{i=0}^{2} risk_{t-i}$, where $t = 1, 2 \ldots, T$.

(2) To evaluate the risk volatility, we calculate the change in three monthly country risk levels $\Delta C_t^{risk} = C_t^{risk} - C_{t-3}^{risk}$, where $t = 4, 5 \ldots, T$.

(3) Considering the variation in risk volatility among African countries, we assess the anchoring effect by calculating the moving average (m) and moving standard deviation (s) over the past three years (36 months).

(4) To identify risk shocks, we establish two conditions. Firstly, the starting node and deadline of risk shocks must be met $\Delta C_t^{risk} - m \geq s$ and $\Delta C_t^{risk} - m \prec s$ respectively. Second, risk shocks must last for at least two periods and meet $\Delta C_t^{risk} - m \geq 2s$ at least one period.

(5) Due to the higher frequency of country risk fluctuations in Africa, we have adjusted the observation time window parameter to 3 months to more accurately detect short-term abnormal risk fluctuations. Additionally, the calculation of anchored risk levels in each country is based on a 36-month time window to distinguish between different periods. In Section 5.4, sensitivity analyses are conducted on these two time parameters.

(6) Based on the obtained results of risk shocks (Fig. 3), South Africa, as China 's most significant African trading partner, consistently maintained a relatively low overall risk level. The classification basis for risk standards provided by the database was added in Fig. 3. The results indicate that South Africa's national risk does not seem to be high in absolute terms; however, there are still occasional abnormal fluctuations which can be considered as risk shocks. In order to align with trade survival data, three indicators are calculated annually for assessing risks shocks in African countries: occurrence of risk shocks (risk), duration of risk shocks (risk_t), and sum of weighted intensities (risk_s);

3.2. Measuring the stability of export relationships with Africa

This study examines the duration of export relationships between Chinese firms and African destinations. The trade data used in this research is sourced from the China Customs Import and Export Database, covering the years 2000 to 2014. The database, collected by China's General Administration of Customs (GAC), provides comprehensive information on firm-level import and export transactions between China and African countries. It includes details such as firm name, product type, amount, quantity, price, country (region), and product code (HS 8-digit). The dataset comprises a total of 10.08 million transaction records of Chinese exporters to various African countries.

To analyze the stability of export relationships, this study constructs survival data based on the duration of a firm's export activities (Breslow, 1974; Zhou et al., 2019). A longer export duration indicates higher export stability for the firm. Since China's exports to African markets exhibit significant heterogeneity, this study also constructs survival data based on firm-country export duration. If a firm temporarily or intermittently exports to a particular destination, it suggests a lack of stability in the firm-destination export relationship. Therefore, the duration of a firm's export activities to a specific country from the beginning of exporting to the suspension of exporting is considered as the firm-country export duration. The suspension of exporting to the target country during the observation period is defined as a failure event.

There are two technical issues to consider when dealing with survival data in this study. Firstly, the survival status of firms from their inception to the beginning of the sample observation period is unknown, creating a statistical left-censoring problem. Secondly, some firms had not exited the market by the end of the observation period in 2014, resulting in right censoring. To address these issues, this paper restricts the sample to the observation within a specific time frame and focuses on newly established firm-country export relationships after 2000. Specifically, We identify the entry time of the firm-country export relationship as the year when the firm first exported to that country between 2000 and 2014. Moreover, we employ survival analysis methodology to enhance estimation efficiency by incorporating termination variables for samples containing right-censored data. For export relationships still active in 2014,

[1] National (regional) economic and political uncertainty indices are derived from the WUI (World Uncertainty Index) database.

[2] The download URL for the original data used in this article is https://www.prsgroup.com/explore-our-products/icrg/.

[3] These countries are Angola, Burkina Faso, Botswana, Cameroon, Democratic Republic of the Congo, Republic of the Congo, Algeria, Egypt, Ethiopia, Gabon, Ghana, Guinea, Gambia, Guinea-Bissau, Kenya, Liberia, Libya, Morocco, Madagascar, Mali, Mozambique, Malawi, Namibia, Niger, Nigeria, Sudan, Senegal, Sierra Leone, Somalia, Togo, Tunisia, Tanzania, Uganda, South Africa, Zambia, and Zimbabwe, respectively.

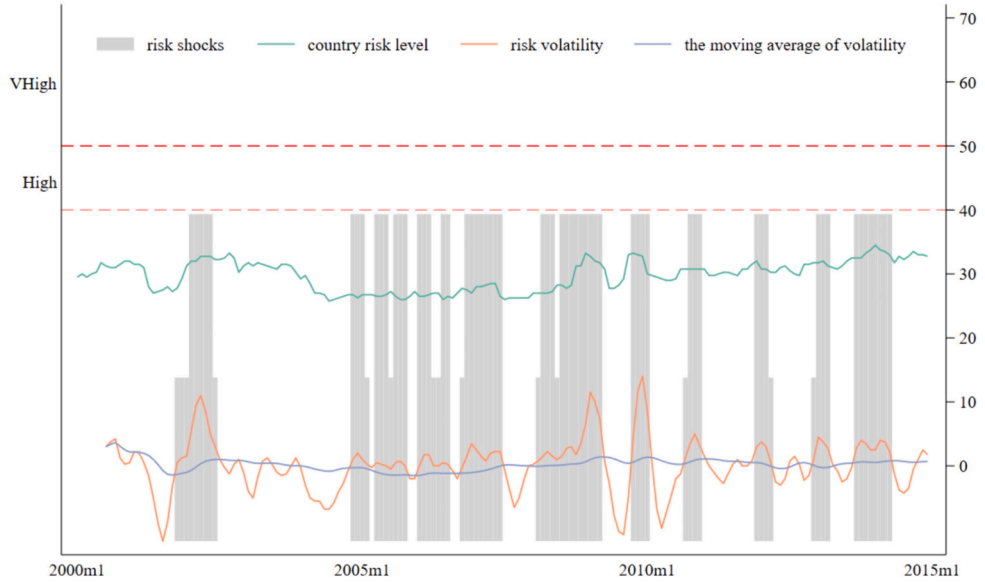


Fig. 3. Measurement of risk shocks in African countries (using South Africa as an example).
Note: (i) The Y-axis on the left side represents the intensity of risk shocks, while the Y-axis on the right side represents the country's risk level. The X-axis represents time (month), ranging from 2000 to 2014. (ii) The green line represents the country's risk level, the yellow color represents the degree of risk volatility, and the blue color represents the average level of risk volatility over different time periods. (iii) The grey squares indicate the occurrence of risk shocks in the current month and their intensity, with longer squares representing higher intensity risk mutations. (iv) We perform reverse indicator processing on monthly risk indicator data for easier understanding, with higher values indicating greater risk. Official standards classify a value of 40–50 as high risk, and anything above 50 as very-high risk. (v) The results presented in this figure were computed using Stata 16 software. Alternatively, readers can also replicate the calculations using other statistical software such as R or Python by following the outlined steps.

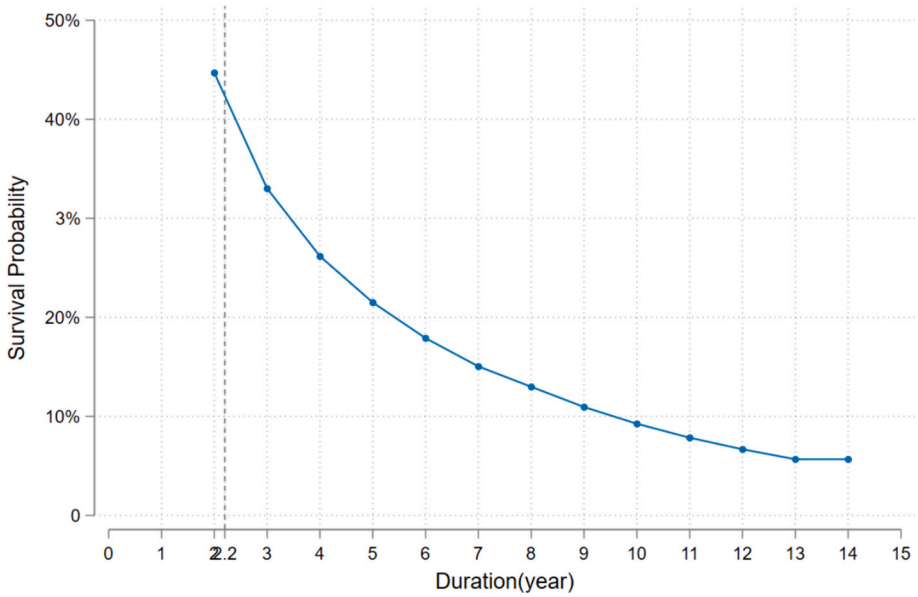


Fig. 4. Life Table of Chinese Firms' Export Relationships with Africa.
Note: (i) The graph depicts the relationship between the length of survival (X-axis) and the probability of survival of the export relationship (Y-axis). (ii) The life table used for this analysis excludes data from export relationships that have occurred only once.

we assign a termination variable value of 1, whereas for exporting markets, an exit variable value of 0 is assigned. Consequently, the maximum theoretical duration for a single relationship is set at 13 periods.

We analyzed the distributional characteristics of firms' export relations between China and Africa in terms of survival durations. In

Fig. 4, the survival curve demonstrates a clear negative duration dependence indicating that the export relationship's survival rate tends to decline over time before stabilizing. Phenomenon can be attributed to the heightened uncertainty shocks faced by firms during the initial stage of exporting. Additionally, more than half of the firm-country export relationships only endure for one year. However, if an export relationship surpasses two years, the risk of exiting the export market significantly reduces (Geishecker et al., 2019). Our findings reveal a less optimistic outlook on the longevity of Chinese firms in African markets based on the duration of their export relationships. Fig. 4 illustrates that Chinese enterprises' average duration for export relationships with African countries is merely 2.2 years old; less than 20 % of enterprise-country-level export relationships survive for more than five years, and less than 10 % survive for more than 10 years.

Furthermore, this study aims to analyze the impact of risk shocks in Africa on the stability of Chinese firm exports to Africa. Additionally, we examine whether the export relationships is affected by risk shocks at the destination, and further explore the differences in survival rates within the subsample. Comparing the results in Fig. 5, it can be observed that export relationships encountering risk shocks exhibit a lower overall survival rate, with mean survival times of 2.1 and 2.4 years for the two sub-samples respectively. These findings suggest that risk shocks in African countries may decrease the duration of export relationships between firms and countries.

4. Model construction and data description

4.1. Survival analysis method

In this paper, we utilize the Kaplan-Meier product-limit method to estimate the cumulative survival and hazard functions of firms. Eq. 1 represents the probability that the duration of a trade relationship i exceeds t year. Further, eq. 2 shows the nonparametric estimation of the survival function, n_k denotes the number of export relationships that are at risk during k period, d_k denotes the number of export relationships observed during the same period, and $n_k - d_k$ denotes the number of export relationships that are surviving during the same period.

$$S_i(t) = pr(T_i \succ t) \tag{1}$$

$$\widehat{s}(t) = \prod_{k=1}^{t} \frac{n_k - d_k}{n_k} \tag{2}$$

The hazard function $h_i(t)$ is expressed as the probability of a firm's export relationship exiting the market in t period under the condition that the relationship survives in $t-1$ period, as shown in eq. 3, and the nonparametric estimation of the hazard function is shown in eq. 4.

$$h_i(t) = pr(t-1 \prec T_i \leq t | T_i \succ t-1) = \frac{pr(t-1 \prec T_i \leq t)}{pr(T_i \succ t-1)} \tag{3}$$

$$\widehat{h} = \frac{d_k}{n_k} \tag{4}$$

According to previous studies, the Cox proportional hazards model is employed in this paper. One of its significant advantages is that it does not require an explicit definition of the benchmark hazard function. The Cox model is considered a semi-parametric model, where the hazard rate of the continuous-time model is represented as a product of two components. The first component represents the benchmark hazard function, indicating the hazard rate when all covariate values are zero. The second component comprises a weighted linear combination of covariates, quantifying their effects on the hazard function. However, it should be noted that certain assumptions must be satisfied by this linear combination of covariates. Therefore, we construct the following Cox model for our research.

$$h(t,X) = h_0(t) exp\left(\beta_1 risk_i + \beta_2 C_{ip} + \alpha_z\right) \tag{5}$$

Additionally, Eq.(5) can be rewritten by taking logarithms on both sides of Eq. (5) as Eq.(6).

$$logh(t,X) = logh_0(t) + \beta_1 risk_p + \beta_2 C_{ip} + \alpha_z \tag{6}$$

where $h(t,X)$ means the hazard rate of the firm-country export relationship after taking into account the effect of covariates. There is a conditional probability of surviving the export market longer than the $t-1$ period and exiting at the t period. In addition, $h_0(t)$ means the baseline risk function, which reflects the common characteristics of the sample and only depends on time. The exponential term is a combination of covariates, which mainly include the indicators of risk shock and other covariates affecting survival.

4.2. Data source and descriptive statistics

This paper examines the influence of risk shocks on the duration of firm-country export relationships. The primary variable of interest is the risk shock in African countries, as measured in Section 3. We utilize risk as the key indicator in our benchmark regressions and further test its robustness using risk_t and risk_s. The raw data for calculating the risk shock indicators were sourced from

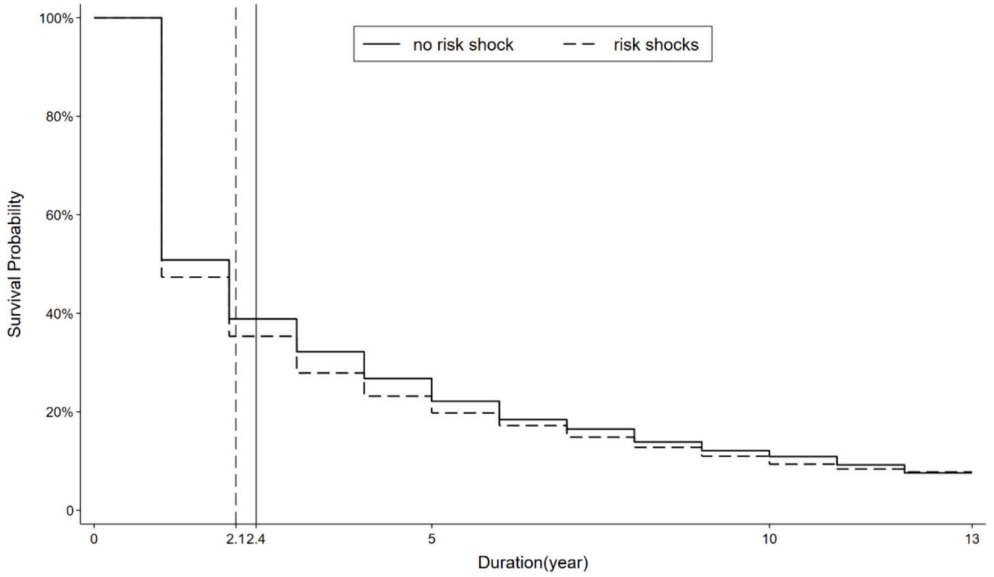


Fig. 5. Comparison of Survival Functions for Whether a Risk Shock Occurs.
Note: (i) This figure is based on the survival data for exports to Africa from 2000 to 2014 collected from China's General Administration of Customs. (ii) The paper demonstrates the disparity in survival status between export relationships that were exposed to risky shocks and those that were not exposed to such shocks.

the ICRG database.

Following the existing literature, this study incorporates additional covariates that influence the survival of export relationships. The covariates are defined as follows: (1) Firm size (size): Total exports as a proxy for firm size, given that higher total exports generally indicate larger firms. (2) Intensity of the export relationship (intensity): The total exports of the firm to specific African countries are calculated to measure the intensity of the export relationship. (3) Experience of exporting to Africa (exper): This variable captures whether the enterprise has previously exported to other African countries prior to establishing new export relationships. A more extensive experience of exporting to Africa is expected to contribute to longer trade relationships with the continent. The firm size, export intensity, and export experience are derived from data collected from China's General Administration of Customs. (4) Bilateral political relations (degree): China's partnership diplomacy is a continuous, comprehensive, and dynamic diplomatic strategy for the 21st century (Wilde & Xing, 2021). The degree of partnership diplomacy relations is used as a measure of bilateral political relations.[4] (5) Economic development (light). The relationship between the level of economic development and the market size of export relationships in African countries is often measured by indicators such as Gross Domestic Product (GDP). However, there are concerns about the reliability of these indicators, especially in countries with less developed economies and higher country risks. Donaldson & Storeygard (2016) argue that regions with higher levels of economic development tend to have higher degrees of urbanization and greater use of nighttime lighting. Therefore, this study uses the intensity of nighttime lights as a measure of annual economic development in African countries, which helps to mitigate the impact of data measurement errors. The study also includes fixed effects at the regional and annual levels for more accurate analysis. Table 1 presents the descriptive statistics of the main variables.

4.3. *Proportional hazards assumption test*

The Cox proportional hazards model offer the advantageous of its flexible setting. However, an additional crucial aspect of this model is the assumption of proportional hazards, which necessitates meeting certain criteria before conducting survival analysis. This assumption states that the risk rate of different types of covariates follows a fixed proportion of change and does not vary over time. The risk rate is determined by comparing the hazard between two observed individuals, as shown in the formula. In the formula, X and X* represent different values of the covariates. It is important to note that the risk rate is solely dependent on the values and coefficients of the covariates and is independent of time.

$$HR = \frac{h(t,X)}{h(t,X^*)} = \frac{h_0(t)exp(\sum \beta X)}{h_0(t)exp(\sum \beta X^*)} = exp\left[\sum \beta(X - X^*)\right] \tag{7}$$

[4] Table A1 of the Appendix is available for those interested in the criteria for assigning levels of bilateral partnerships between China and Africa. The documents of partnership diplomatic relations between China and African countries can be found on the website of the Ministry of Foreign Affairs of China.

Table 1
Summary statistics of the main variables.

Variables	Definitions	Mean	Std.Dev	Min	Max	Observations
risk	whether a risk shock occurs	0.66	0.47	0.00	1.00	831,682
risk_t	duration of the risk shock	3.53	3.27	0.00	12.00	831,682
risk_s	the total intensity of risk shock	6.56	6.13	0.00	24.00	831,682
size	firm size	12.60	2.09	0.69	20.56	831,682
intensity	export intensity	10.29	1.77	5.58	15.32	816,291
exper	export experience with Africa	0.51	0.50	0.00	1.00	831,682
degree	bilateral political relations	1.59	0.80	0.00	3.00	831,682
light	economic development	11.61	1.99	3.75	14.83	831,682

Note: (i) Firm size (size), export intensity (intensity), and economic development (light) are expressed in logs. (ii) export intensity (intensity) is truncated at 1 % and 99 % to exclude extreme observations.

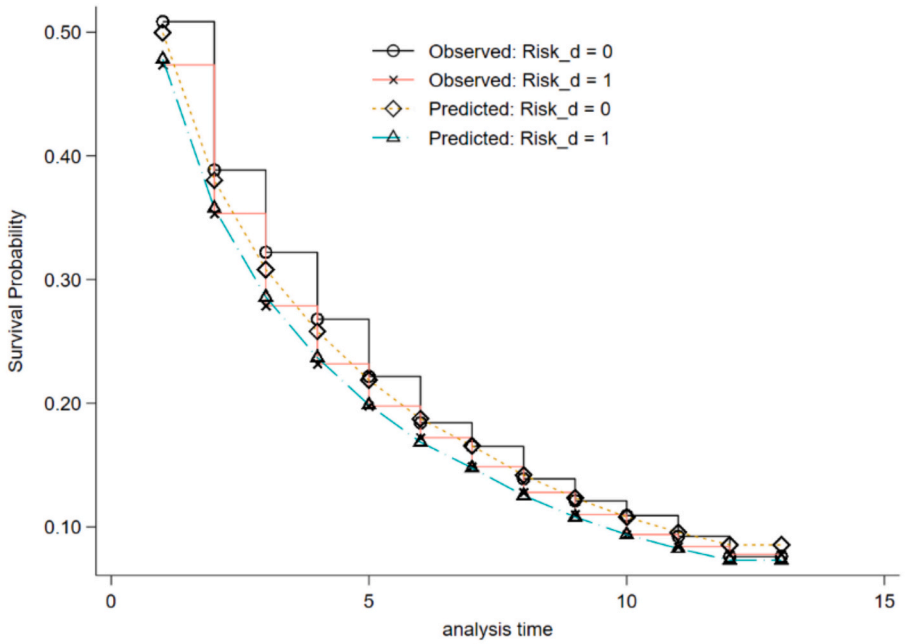


Fig. 6. Test Results for the Proportional Hazards Assumption (Graphical Evaluation method).

This paper employs is employed in this paper to test the proportional risk hypothesis. Fig. 6 presents the test results for the assumption of proportional hazards, indicating that the two log-log survival curves exhibit approximate parallelism. This observation suggests that it is justifiable to utilize the Cox proportional hazards model.

5. Empirical analysis

5.1. Benchmark results

The estimation results of the benchmark regression in this paper are presented in columns 1–3 of Table 2. In column 1, the regression only includes the variable of risk shocks as an explanatory variable. The estimated coefficient is positive and statistically significant, indicating a positive correlation between the survival risk of the firm-country export relationships and the risk shocks from the destination. Columns 2 and 3 include additional covariates and fixed effects in the estimation. The estimated coefficients on the risk shocks variable remain positive and statistically significant at the 1 % level, suggesting that risk shocks significantly increase the survival risk of the firm-country export relationship. In other words, after accounting for other covariates, local risk shocks significantly reduce the probability of survival and duration of export relationships, supporting Hypothesis 1.

Next, we examine the lagged and expected effects of risk shocks on export relationships. The lagged effect refers to the potential delay in the negative impact of risk shocks due to information transmission costs. Additionally, firms may adjust their export decisions based on their assessment of the likelihood of risk shocks in the destination country, which can influence the duration of the export relationships. To investigate these effects, the model includes lagged risk shocks from one year ago (L1_risk) and expected risk shocks from one year ahead (F1_risk). The lagged effect and expected effect are presented separately in columns 4 and 5 of Table 2. The

Table 2
Effect of risk shocks on firm-country trade relations survival.

	(1)	(2)	(3)	(4)	(5)	(6)
	Benchmark			lagging effect	predicted effect	lasting effect
risk	0.062***	0.089***	0.033***	0.031***	0.025***	0.033***
	(29.97)	(43.87)	(15.33)	(14.24)	(11.54)	(15.44)
L1_risk				0.013***		0.010***
				(6.26)		(4.81)
L2_risk						0.024***
						(11.10)
F1_risk					0.045***	
					(20.91)	
size		−0.104***	−0.101***	−0.101***	−0.101***	−0.101***
		(−173.44)	(−173.64)	(−173.74)	(−174.04)	(−173.88)
intensity		−0.004***	0.002**	0.002***	0.002***	0.002***
		(−5.69)	(2.49)	(2.68)	(3.06)	(2.94)
exper		−0.040***	−0.040***	−0.040***	−0.040***	−0.040***
		(−20.24)	(−21.15)	(−21.20)	(−21.32)	(−21.33)
degree		0.007***	0.004***	0.004***	0.004***	0.004***
		(77.23)	(32.92)	(32.71)	(31.76)	(32.36)
light		−0.014***	−0.013***	−0.013***	−0.013***	−0.013***
		(−138.49)	(−107.20)	(−107.31)	(−109.10)	(−107.53)
Region FE			YES	YES	YES	YES
Year FE			YES	YES	YES	YES
N	831,682	816,291	816,291	816,291	734,966	816,291
Likelihood	−8,230,271	−8,039,550	−7,990,683	−7,990,673	−7,990,572	−7,990,642

Note: Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

findings reveal that risk shocks have a delayed negative impact on export relationships survival. Notably, there is also a significantly positive and stronger expected effect of destination risk shocks on the probability of export relationship survival. In addition to direct trade losses resulting from suspension or interruption of export relationships with Africa due to risk shocks, there is also a more substantial impact from perceived risk shocks. Many exporters may lower their return expectations and adopt a wait-and-see approach due to concerns about local risk shocks and export failures in African markets, leading to incalculable indirect losses in China-Africa trade cooperation. Furthermore, the study examines the lasting effect by including lagged risk shocks from one year ago (L1_risk) and two years ago (L2_risk). Column 6 indicates that the negative impact of risk shocks on the duration of export relationship survival may persist.

5.2. Heterogeneity analysis

5.2.1. Sources of risk shocks

The main sources of risk for African countries include terrorism, economic crises, regional conflicts, and regime change. This study examines the heterogeneous effects of different risk shocks on the survival of export relationships. We classify these risk shocks into two categories: economic risk shock (erisk) and political risk shock (prisk). The results, considering both types of risk shocks, are presented in columns 1 and 2 of Table 4. While the coefficient for political risk shock is 0.040, it is evident that both types of risks diminish the duration of a firm's export relationships; however, the negative impact from political risk shock is more pronounced. Political risk shocks may occur less frequently but have a higher intensity of impact in Africa compared to economic risk shocks. Furthermore, political risk shocks are often accompanied by economic risk shocks, but economic risk shocks do not necessarily lead to political risk shocks.

5.2.2. Firm ownership and trade style

This study investigates how risk shocks affect export relationships among firms with diverse ownership structures and trading methods. To analyze this heterogeneity, we introduce interaction terms that capture the interplay between risk shocks and dummy variables representing different forms of ownership and trading modes. We classify firms into three distinct categories based on their

Table 3
The definitions, classifications, and distribution statistics of the types of firm ownership.

Definition	soe =1		private =1	foreign = 1	
Classification	state-owned firms	collective firms	private firms	foreign-owned firms	Sino-foreign cooperative firms
Proportion	11.8 %	4.8 %	48.8 %	19.6 %	15.0 %

Note: (i) The original data on firm ownership is derived from the China Customs database. (ii) This study categorizes the similarity of enterprise ownership characteristics into three main categories, and further investigates the impact of state-owned background and foreign capital background. (iii) The results indicate that private enterprises form the majority of the research sample.

Table 4
The results of firm-level heterogeneity analysis regression.

	(1)	(2)	(3)	(4)	(5)
	Economic risk shocks	Political risk shocks	Ownership type	Foreign background	Trades tyle
risk			0.038***	0.032***	0.044***
			(9.88)	(8.32)	(14.00)
erisk	0.025***				
	(11.77)				
prisk		0.040***			
		(20.13)			
risk*soe			−0.013*		
			(−1.73)		
soe			0.065***		
			(10.63)		
risk*foreign				−0.011*	
				(−1.66)	
foreign				−0.056***	
				(−10.53)	
risk*style					−0.041***
					(−7.64)
style					0.036***
					(5.80)
Controls	YES	YES	YES	YES	YES
RegionFE	YES	YES	YES	YES	YES
YearFE	YES	YES	YES	YES	YES
N	816,291	816,291	265,134	278,559	559,770
Likelihood	−7,990,708	−7,990,637	−2,658,607	−2,803,796	−5,342,461

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

ownership characteristics as outlined in Table 3, which also provides statistical data regarding the distribution of these ownership types. Additionally, we specifically focus on two styles of trading - general trade (style = 1) and processing trade (style = 0).

The regression results are presented in columns 3–5 of Table 4. Column 3 demonstrates the differential impact of risk shocks on the stability of export trade relations between state-owned firms and private firms. The coefficient of the interaction term risk*soe is significantly negative, indicating that state-owned firms are less affected by risk shocks compared to private firms. This implies that state-owned firms may have a stronger ability to withstand the impact of risk shocks. However, it is important to note that risk shocks have a a more pronounced influence on the stability of export relationships for private firms. Additionally, it is noteworthy that the majority of our exporters to Africa are private firms,with their market share expected to continue expanding in the future.

Some studies have found that firms with foreign capital have more efficient access to market information, resulting in a higher probability of risk avoidance (Ilmakunnas & Nurmi, 2010). To account for the foreign background of the firm, it is included as a variable in column 4. The results indicate a significantly negative coefficient for the interaction term risk*foreign, indicating that firms with a foreign background experience less negative impacts of risk shocks on their export relationships. This implies that export relationships with a foreign background exhibit relatively greater stability when confronted with risk shocks. Furthermore, column 5 presents the results of a heterogeneity analysis on trade style. The significantly negative coefficient of the interaction term risk*style indicates that the stability of export relationships in processing trade is more adversely affected by risk shocks. This can be attributed to processing trade firms relying on raw materials and order sources that are more susceptible to fluctuations in market country risk. As a

Table 5
The results of product-level heterogeneity analysis regression.

	(1)	(2)	(3)	(4)
	Top 3 categories			
	textile products	metal products	electromechanical equipment	Non-resource products
risk	0.033***	0.025***	0.047***	0.052***
	(7.89)	(4.41)	(8.34)	(11.00)
risk_non_resource				−0.025***
				(−5.36)
non_resource				−0.044***
				(−12.58)
Controls	YES	YES	YES	YES
Region FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	131,721	99,987	177,491	816,291

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

result, processing export relationships are more sensitive to uncertainties in supply and demand information. In contrast, general trade firms can actively maintain trade relations with the destination through strategies such as delayed exporting, product upgrades, and multi-market combinations.

5.2.3. Types of export products

This study delves deeper into the varied impacts of African country risk shocks on export trade relations across different product categories. Analyzing China's exports to Africa from 2002 to 2014, the top three product groups were textile raw materials and products, base metals and products, and electromechanical equipment and parts. Subsequently, a sub-sample regression was conducted on these product categories, with the results presented in columns 1–3 of Table 5. The findings indicate a significant negative influence of African country risk shocks on the stability of export relationships, particularly affecting high-unit-price products like electromechanical equipment. Furthermore, the classification of export products into resource and non-resource categories revealed that non-resource products exhibit greater resilience to risk shocks in columns 4 of Table 5. Non-resource products demonstrate higher resilience to risks compared to resource products, aligning with existing research findings that homogeneous goods have a higher hazard rate than differentiated products (Besedeš & Prusa, 2006). Moreover, the resilience can be linked to the industrialization efforts in African nations and the significant infrastructure support from China, which has bolstered the demand for non-resource products. The emerging trend indicates a growing reliance on non-resource products driven by ongoing industrialization and infrastructure projects in African countries, potentially resulting in a trade diversion effect with resource products.

5.3. Robustness checks

Three robustness checks are conducted to verify the basic results of this study. Firstly, two different indicators of risk shocks are used: risk_t, which measures the total duration of risk shocks in the current year, and risk_s, which measures the weighted intensity score of risk shocks in the current year. The results presented in columns 1 and 2 of Table 6 respectively demonstrate that both indicators exhibit significantly positive coefficients for the risk shock variables at a level of significance of 1 %. This indicates that risk shocks increase the survival risk and reduce the duration of firm-country export relationships. Secondly, we re-estimate the association between risk shocks and export stability using the Weibull risk model (Bousquet et al., 2006). The results displayed in column 2 indicate that substituting research methods does not alter either sign or significance levels for the coefficient associated with the risk shock variable. Thirdly, to address the potential influence from global risk events, such as the 2008 financial crisis, on Chinese firms' export decisions and export viability in Africa, we exclude the subsample of trade relations that entered the export market in 2008. The results depicted in column 4 demonstrate that even after excluding these cases, coefficients related to risk shock remain significantly positive, further confirming the robustness of the benchmark regression results.

5.4. Sensitivity analysis

The utilization of two distinct time windows in the calculation of abnormal fluctuations in risk within African countries is pivotal for effectively identifying risk shocks. Given the short-term nature of risk fluctuations in African countries, a shorter observation time window of 3 months is set to better detect abnormal risk fluctuations. Additionally, to emphasize differences between countries, the time window for calculating each country's anchored risk level is set at 36 months. To further explore the impact of these parameters on the empirical results, a sensitivity analysis is conducted by gradually extending the time window.

The regression results in Table 7 show that as the observation time window expands, the coefficients of the core explanatory variables decrease or become insignificant. This phenomenon arises due to the failure to recognize abnormal fluctuations in risk, such as multiple short cycles, as the window period extends. Consequently, this leads to a diminished validity of the core explanatory variables. Furthermore, findings from columns 4–6 indicate that while there is some impact on coefficient magnitudes resulting from

Table 6
The results of robustness test.

	(1)	(2)	(3)	(4)
	replace the core indicators		replace regression methods	eliminate special event
risk			0.017***	0.017***
			(9.53)	(7.34)
risk_t	0.004***			
	(13.63)			
risk_s		0.002***		
		(12.37)		
Controls	YES	YES	YES	YES
Region FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	816,291	816,291	816,291	730,778
Likelihood	−7,990,694	−7,990,703	−1,065,411	−6,989,282

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 7
The results of parametric sensitivity analyses for observation time windows and anchoring risk.

	(1)	(2)	(3)	(4)	(5)	(6)
	observation time window			anchoring risks		
	3 months	6 months	9 months	3 years	4 years	5 years
Risk_d	0.033***	0.022***	−0.008	0.033***	0.013***	0.024***
	(17.05)	(5.42)	(−2.47)	(17.05)	(7.35)	(14.86)
Controls	YES	YES	YES	YES	YES	YES
Region FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
N	816,291	816,291	816,291	816,291	816,291	816,291

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

different choices of time windows for calculating anchored risk levels, these variations do not affect the overall conclusions drawn in this paper. Therefore, through sensitivity analysis it can be confirmed that the selected parameter values are reasonable and adjustments made within an acceptable range do not alter the conclusions.

In addition, to further validate the risk shock data, a set of simulated data was randomly generated for comparison with the actual data. The distribution characteristics of both datasets were kept consistent. The empirical results, presented in Table 8, indicate that the key explanatory variables in the randomly generated simulated data are not statistically significant. This further confirms the reliability of the risk shock data calculated in this study.

6. Expanding the analysis: The moderating role of export strategy

6.1. Market strategy

6.1.1. Market selection strategy

This study focuses on the export relationship between a single firm and a destination country. When the number of firms exporting to the same destination increases, it may divert resources from enterprise exports, particularly in uncertain destinations (Héricourt & Nedoncelle, 2018). To analyze the stability of export relationships under risk shocks, this paper calculates the number of destinations (markets) to which firms export to Africa each year. The study also examines how firms' market quantity moderates this effect by introducing an interaction term (risk*markets). The results presented in column 1 of Table 9 show a significantly positive coefficient for the interaction term, indicating that an increase in the number of export destinations exacerbates the negative impact of risk shocks on firms' export relationship stability. Furthermore, this study investigates differences in this effect between state-owned firms and private firms. Columns 2 and 3 reveal a positive coefficient for markets, supporting the idea that a multi-market strategy reduces the risk resistance of trade relations and shortens their duration. Additionally, the coefficient of the interaction term markets *soe is significantly negative, suggesting that survival risks are more pronounced in multi-markets and have a greater impact on private firms' export relationships.

Gullstrand & Persson (2015) found that there exists a disparity in the sunk costs and expected revenues of firms between core and non-core export markets. As a result, firms tend to remain in core markets while entering and exiting non-core markets frequently. This study aims to explore whether there is intergroup heterogeneity in the impact of destination risk shocks on the export relationships between core and non-core markets. To achieve this, we identified the firm's current year's core market for exports (core). We assigned a value of 1 to core if it represented the firm's export relationship with the highest exports in the current year, otherwise it was assigned a value of 0. Subsequently, we introduced an interaction term (risk*core) into our model, and the results are presented in column 4 of Table 6. The coefficient of the interaction term is significantly positive, suggesting that the export relationships in the core market are more resilient to risks compared to those in the non-core market. Additionally, column 5 presents the results of the subsample regression, which indicate that the export relationships in the core market are less negatively affected by the risk shock. Furthermore, in column 6, we examine the heterogeneity of the types of firm ownership. The coefficient for core*soe is significantly positive, implying that private firms have become more effective in enhancing the stability of export relations by focusing on core markets.

6.1.2. Learning strategies: Learning by exporting and learning by neighboring

The learning-by-exporting theory suggests that previous exporting activities, which provide knowledge of overseas market operation experience, market demand, and consumer preferences, can enhance firms' productivity. This, in turn, increases the size and viability of future exporting activities. Esteve-Pérez et al., 2013 argue that the information spillover effect is particularly important for improving the survival of export relationships with high-risk countries. Therefore, this paper considers the number of years in which the enterprise has exported as a measure of the firms' level of exporting experience (exp_t). Table 10, column 1 presents the effect of the learning-by-exporting strategy, demonstrating that it can mitigate the negative impact of risk shocks on export relationships.

According to Fernandes & Tang (2014), exporters can gain market information not only from their own exporting experience but also from neighboring firms. Export agglomeration, as an informal channel for information dissemination, helps reduce the information search cost for exporting firms. Additionally, export spillovers positively influence the decision-making process and survival

Table 8
Comparison of empirical results from random data and real data.

	(1)	(2)
	real	random
risk	0.033***	
	(17.05)	
risk_r		0.001
		(−0.83)
Controls	YES	YES
Region FE	YES	YES
Year FE	YES	YES
N	816,291	816,291

Note: (i)The control variables in this table are consistent with the baseline regression. (ii)Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 9
Effects of market selection strategies and export relationship risk resilience.

	(1)	(2)	(3)	(4)	(5)	(6)
	multi-market strategy			core market		
	full-sample	sub-sample		full-sample	sub-sample	
risk	0.006			0.041***		
	(1.32)			(16.15)		
risk*markets	0.032***					
	(6.65)					
markets	0.206***	0.218***	0.201***			
	(48.58)	(65.03)	(38.23)			
markets *soe			−0.131***			
			(−11.42)			
risk*core				−0.030***		
				(−7.23)		
core				−0.298***	−0.311***	−0.268***
				(−80.00)	(−98.83)	(−53.07)
core*soe						0.092***
						(9.40)
soe			0.187***			0.050***
			(18.04)			(8.49)
Controls	YES	YES	YES	YES	YES	YES
Region FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
N	816,291	536,342	208,557	816,291	536,342	208,557
Likelihood	−7,988,832	−5,114,033	−2,035,252	−7,986,508	−5,112,526	−2,034,839

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 10
Effects of learning strategies on export relationships risk resilience.

	(1)	(2)	(3)
	learning by exporting	learning by neighboring	
risk*exper_t	−0.022***		
	(−39.64)		
risk*f_num		−0.179***	
		(−125.24)	
risk*sf_num			−0.173***
			(−122.69)
Controls	YES	YES	YES
Region FE	YES	YES	YES
Year FE	YES	YES	YES
N	816,291	816,291	816,291
Likelihood	−7,912,511	−7,911,435	−7,911,519

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

rate of nearby firms in initiating exports (Hu & Tan, 2016). This study defines the number of Chinese firms exporting to a specific country in a given year (f_num) as a proxy for the export signals from neighboring firms. In other words, a higher number of firms exporting to a specific country in the same year indicates stronger export signals from neighboring firms. Furthermore, firms can also learn from those that already have experience in exporting to a specific destination. The cumulative number of Chinese firms exporting to a specific African country (sf_num) is used in this paper to measure the convenience of obtaining information from firms with historical experience. Columns (2) and (3) demonstrate the effects of the learning by neighboring strategies, showing that these strategies can mitigate the negative impact of risk shocks on export relationships.

6.1.3. Waiting strategy

Facing increasing market uncertainty in the destination, many firms may temporarily reduce the frequency of exports or even halt exports altogether to actively mitigate potential losses from risk shocks. They wait for a certain period before re-exporting to the country. This paper defines the waiting strategy (wait) as the intermittent exporting behavior of firms for specific markets. If the firm-country export relationship aligns with the definition of the wait strategy, we assign wait = 1; otherwise, wait = 0. After analysis, we find that 132,762 export relationships (15.96 %) are identified as wait = 1. We then introduce the waiting strategy variable and its interaction term with risk shocks to examine the effectiveness of the waiting strategy in enhancing the risk resilience of firms' export relationships. The results are presented in column 1 of Table 8. The coefficient of the interaction term is significantly negative, indicating that the waiting strategy can mitigate the negative impact of risk shocks on export stability.

In addition, for export relationships involving the waiting strategy, a crucial question arises: what is the optimal waiting time? Therefore, this paper further investigates the optimal waiting time by examining the marginal effect of waiting duration (wait_n). The number of relationships with a waiting duration of 1 year, 2 years, and 3 years are 84,999 (10.41 %), 32,397 (3.97 %), and 16,377 (2.01 %) respectively.[5]

There may be a nonlinear relationship between the choice of waiting time and export stability. To further examine the moderating effect of waiting duration, we include additional variables such as the waiting duration (wait_n), its quadratic term (wait_n2), and the interaction terms with risk shocks (risk*wait_n and risk*wait_n2). In column 2 of Table 11, the results indicate that the coefficient of risk*wait_n is significantly negative, while the coefficient of risk*wait_n2 is significantly positive. This suggests that the negative impact of risk shocks on export stability is influenced by the waiting duration. Fig. 7 illustrates that the margin effect of waiting time has a nonlinear characteristic. When the margin effect is negative, it means that the waiting strategy can mitigate the negative impact of risk shocks on export stability. However, this effect follows an inverted U-shape, gradually strengthening over time initially and then weakening after the optimal timing of re-export (T). Additionally, when the waiting duration exceeds the threshold (D), the margin effect becomes positive, indicating that the waiting strategy exacerbates the negative impact of risk shocks on export stability. Therefore, we further investigate the effects of waiting strategy and waiting time on export stability in the subsample affected by risk shocks. The results, presented in columns 3 and 4, are consistent with the interaction term model.

Therefore, we find that the impact of firms' waiting strategy on risk resistance depends on the timing of firms' re-export choices. On one hand, the waiting strategy allows firms to avoid the persistent impact of risk shocks by temporarily halting exports. Additionally, the probability of successfully re-exporting is increased as firms learn from the experience of early adopters during the waiting period. On the other hand, the waiting strategy comes with the opportunity cost of forgoing potential trade profits. In summary, the overall effect of the waiting strategy on risk resistance is determined by the waiting value and opportunity cost during the waiting duration. When the waiting duration is short, the waiting value increases over time, leading to a continuously enhanced net effect. However, as the waiting duration prolongs, the waiting value of risk avoidance and learning effect decreases, while the opportunity cost of the waiting strategy increases. Consequently, the net effect weakens over time, and an excessively long waiting duration may even reduce resilience to risk shocks. In addition to potential trade profit loss and high continuous learning costs, firms may also face market share saturation in the destination and a decline in product competitiveness.

6.2. Non-market strategies

Non-market strategy refers to a strategic approach adopted by firms in non-market environments to improve their overall behavior. One way firms achieve this is by influencing government trade policy-making through lobbying, which creates a favorable policy environment, improves the image and reputation, and enhances the legitimacy of their exports to specific market countries (Khan et al., 2023; McGuire et al., 2012; Yeo & Lee, 2009). In the context of export relationships with Africa, non-market strategies are commonly employed due to the limited degree of marketization. Some star firms can establish direct contact with government officials and local leaders in destination countries, either directly or through their agents (Ciliberto & Jkel, 2021). However, diplomatic representatives can also play a significant role as intermediaries for bilateral firms (Visser, 2019).

There are two channels through which firms can leverage government diplomacy to develop non-market strategies for export destinations. First, minority pioneer firms, acting as industry representatives, can accompany visits by senior officials to express policy demands. Second, policy demands will be widely collected and organized through industry association seminars, entrepreneurial forums, and government symposiums, which are then converted into diplomatic negotiations between bilateral governments on related topics. Therefore, an important form of firms' non-market strategy is presented as a three-dimensional bilateral diplomatic

[5] Figure A1 of appendix illustrates the distribution of waiting durations.

Table 11
Effects of the wait strategy on export relationship risk-resistant capacity.

	(1)	(2)	(3)	(4)
	interaction term model		subsample regression	
	subgroup	moderating effect	subgroup	marginal effect
risk	0.073***	0.049***		
	(33.01)	(22.36)		
risk*wait	−0.067***			
	(−10.92)			
wait	−1.318***		−1.444***	
	(−256.27)		(−307.07)	
risk*wait_n		−0.069***		
		(−11.46)		
risk*wait_n2		0.007***		
		(5.96)		
wait_n		−0.847***		−0.917***
		(−175.20)		(−219.71)
wait_n2		0.078***		0.086***
		(81.27)		(108.63)
Controls	YES	YES	YES	YES
Region FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	816,291	816,291	536,342	536,342
Likelihood	−7,978,174	−7,934,678	−5,075,245	−5,079,906

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

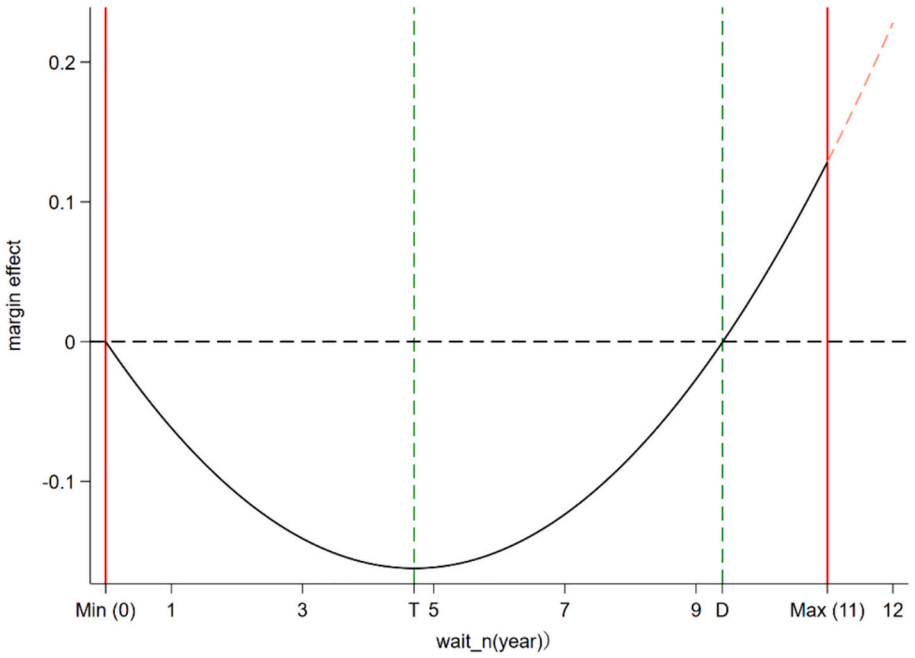


Fig. 7. Margin effect of waiting time on the relationship between risk shocks and export stability.
Note: (i) The graph illustrates the relationship between waiting time (X-axis) and the marginal effect of the waiting strategy on the enterprise's anti-risk ability (Y-axis). (ii) It is important to note that a negative coefficient of the marginal effect indicates that the waiting strategy enhances the enterprise's anti-risk ability (reducing survival risk). This effect demonstrates a non-linear pattern, initially increasing and then decreasing with longer waiting times.

behavior, including government diplomacy, public diplomacy, and civil diplomacy. Due to the difficulty of collecting data on civil diplomacy during the sample period, this paper mainly focuses on the moderating effect of government diplomacy and public diplomacy on the relationship between risk shocks and the stability of export relationships. In our study, the number of head-of-state visits (visit_1) and high-level visits (visit_2) between China and African countries each year is used as a measure of government diplomacy(Ambrocio & Hasan, 2021). Additionally, we measure bilateral public diplomacy by the cumulative data on the establishment

Table 12
Non-market strategies and export relations' risk resilience.

	(1)	(2)	(3)	(4)
	Government diplomacy		Public diplomacy	
	The head of State	High-level visits	Confucius Institute	Friendship Cities
risk*visit_1	−0.017***			
	(−5.99)			
risk*visit_2		−0.002***		
		(−2.66)		
risk*school			−0.106***	
			(−25.42)	
risk*twin				−0.026***
				(−8.67)
Controls	YES	YES	YES	YES
Region FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	784,447	784,447	774,227	816,291
Likelihood	−7,645,954	−7,645,646	−7,487,162	−7,990,233

Note: (i) The control variables in this table are consistent with the baseline regression. (ii) Estimated coefficient z-values in parentheses, ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

of Confucius Institutes and the signing of Friendship Cities in export destinations each year. Table 12 reports the effectiveness of non-market strategies, where each interaction term is significantly negative, indicating that the firm's non-market strategies can promote bilateral diplomacy to enhance the risk resistance of export relationships.

7. Conclusion

The risk shock encountered by Chinese firms in engaging in sustainable trade with African countries presents a substantial obstacle and concern. While existing research predominantly examines the influence of variations among African countries on export decisions, there is a lack of studies that investigate the impact of risk fluctuations in African countries on the longevity of firms' export relationships with Africa and potential countermeasures.

This paper aims to identify risk shocks in African destination countries by the characteristics of fluctuations in risk. It utilizes survival analysis to explore the impact of these risk shocks on the stability of firms' export relations. By analyzing a detailed dataset of Chinese firm-destination export relationships from 2000 to 2014, our findings demonstrate that risk shocks in African destination countries significantly reduce the stability of Chinese firms' export relationships to Africa. Furthermore, private firms and non-foreign firms are more negatively affected by these risk shocks.

Additionally, this paper investigates the moderating role of firms' export strategies in response to risk shocks. With respect to market strategies, our results indicate that learning through exporting, learning from neighboring firms, and adopting a waiting strategy can mitigate the adverse impact of risk shocks on export stability. Notably, the waiting strategy exhibits a non-linear marginal effect. However, the adoption of a multi-destination strategy exacerbates the negative influence of risk shocks on firms' export relationships, particularly for private enterprises. Furthermore, exporters can reduce the negative impact of risk shocks on export stability by adopting non-market strategies, such as promoting bilateral government diplomacy and engaging in public diplomacy behavior.

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Data availability

Data will be made available on request.

Appendix A. Appendix

Table A1
Criteria for Assigning Levels of Bilateral Partnerships between China and Africa.

Diplomatic rhetoric keywords	Assignment Score

(*continued on next page*)

Table A1 (*continued*)

Diplomatic rhetoric keywords	Assignment Score
No diplomatic relations/broken diplomatic relations	0
Diplomatic relations only	1
Friendly cooperation relationship	1.2
(Collaborative) Partnerships	1.5
Comprehensive (cooperative) partnerships	2
Strategic (cooperative) partnerships	2.5
Comprehensive strategic (cooperative) partnership	3

Note: The documents of partnership diplomatic relations between China and African countries can be found on the website of the Ministry of Foreign Affairs of China.

Table A2
Classification Standards of Risk Level in International Country Risk Guide (ICRG) dataset.

Degree	Score
Very Low Risk	80–100
Low Risk	70–79.9
Middle Risk	60–69.9
High Risk	50–59.9
Very High Risk	0–49.9

Note: The International Country Risk Guide (ICRG) model for forecasting financial, economic, and political risk was created in 1980 by the editors of International Reports, a widely respected weekly newsletter on international finance and economics. The ICRG rating comprises 22 variables in three subcategories of risk: political, financial, and economic. A separate index is created for each of the subcategories. The Political Risk index is based on 100 points, Financial Risk on 50 points, and Economic Risk on 50 points. The total points from the three indices are divided by two to produce the weights for inclusion in the composite country risk score. The composite scores, ranging from zero to 100, are then broken into categories from Very Low Risk (80 to 100 points) to Very High Risk (zero to 49.9 points).

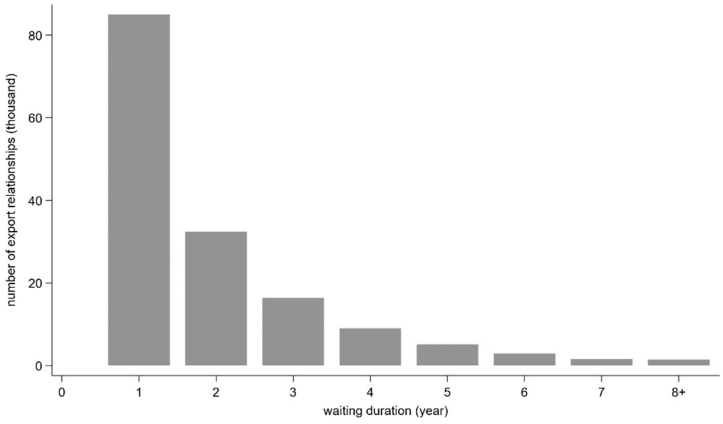


Fig. A1. The distribution of the waiting duration.
Data Source: Author's previous research based on on the survival of Chinese firms' export relationships to Africa.
Note: In order to analyze the data effectively, this paper categorizes firms with a waiting time of more than 8 years, considering their relatively small number.

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